



Basics of Bacterial Genomes & Genome Organization

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Learning Objectives

By the end of this module, participants will be able to:

- Describe chromosome, plasmids, and mobile genetic elements
- Compare genome size, GC content, and gene organization in *E. coli* vs *Klebsiella*
- Explain operons, regulons, and gene density in bacteria
- Interpret GC skew, pan-genome vs core genome, and synteny at a high level
- Recognize AMR gene contexts and virulence loci typical of *E. coli* and *Klebsiella*



Genomes and Genomics

- The word “genome,” coined by German botanist Hans Winkler in 1920, was derived simply by combining **gene** and the final syllable of **chromosome**.
- If not specified, “genome” usually refers to the **nuclear genome**!
- An organism’s **genome** is defined as the complete **haploid genetic complement** of a typical cell.
- The genetic content of the organelles in the cell, is not considered part of the nuclear genome.
- In diploid organisms, sequence variations exist between the two copies of each chromosome present in a cell.
- The genome is the **ultimate source of information** about an organism.

Genomes and Genomics

"**Genes**" are units of genetic information present on the DNA in the chromosomes and chromatin.

" **Genome**" is all the DNA contained in an organism or a cell, which includes the chromosomes plus the DNA in mitochondria (and DNA in the chloroplasts of plant cells).

Importance of studying bacterial genomes

■ Understanding Evolution:

- Genomics provides insights into bacterial evolution, adaptation, and the development of antibiotic resistance and other characteristics.

■ Public Health:

- It helps in tracking bacterial outbreaks, understanding pathogenicity, and identifying antimicrobial resistance genes.

■ Biotechnology Applications:

- Bacterial genomics plays a crucial role in identifying metabolic pathways useful for industrial applications, such as in biofuel production, waste management, and pharmaceuticals.

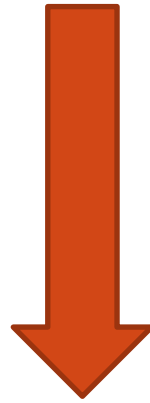
Basic features of bacterial genomes

- The genome of a bacterial cell is primarily composed of a single circular chromosome found in the cytoplasm.
- This chromosome contains the genes and instructions for the survival and reproduction of the cell.
- In addition to chromosomal DNA, the bacterial genome consists of an extra DNA molecule called a plasmid.
- The genome of a bacteria can also include phages and transposons

Prokaryotes and Eukaryotes genome

Prokaryotes	Eukaryotes
Single cell	Single or multi cell
No nucleus	Nucleus
One piece of circular DNA	Chromosomes
No mRNA post transcriptional modification	Exons/Introns splicing

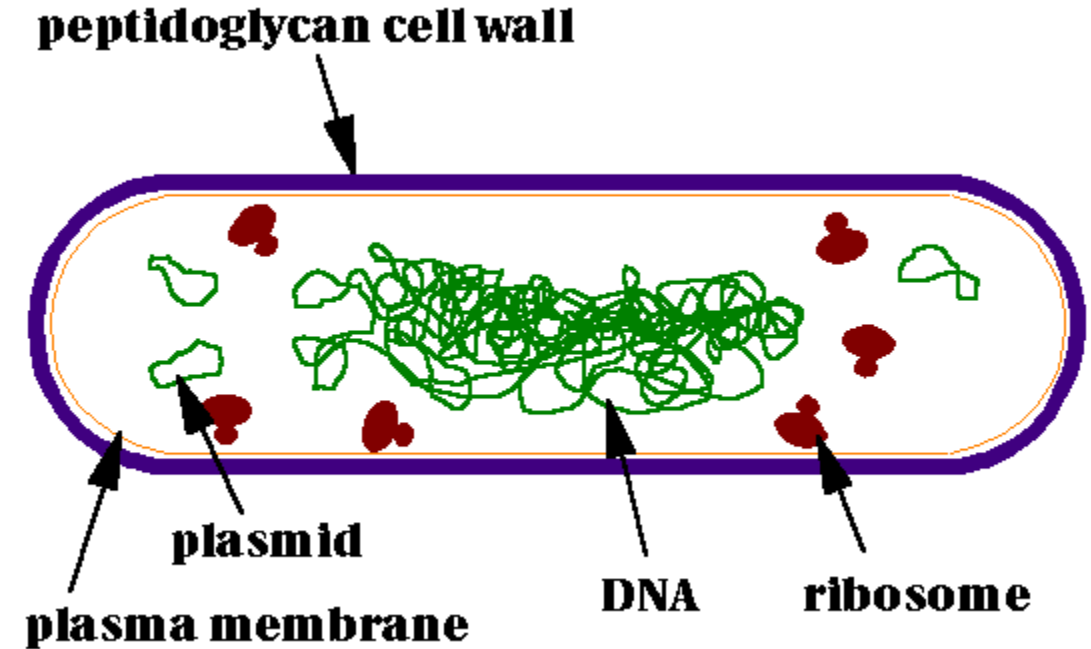
Bacteria are prokaryotes



They lack a nucleus

Prokaryotes

- The genome of *E. coli* contains amount of 4×10^6 base pairs
- > 90% of DNA encode protein
- Lacks a membrane-bound nucleus.
 - Circular DNA and supercoiled domain
- Histones not present

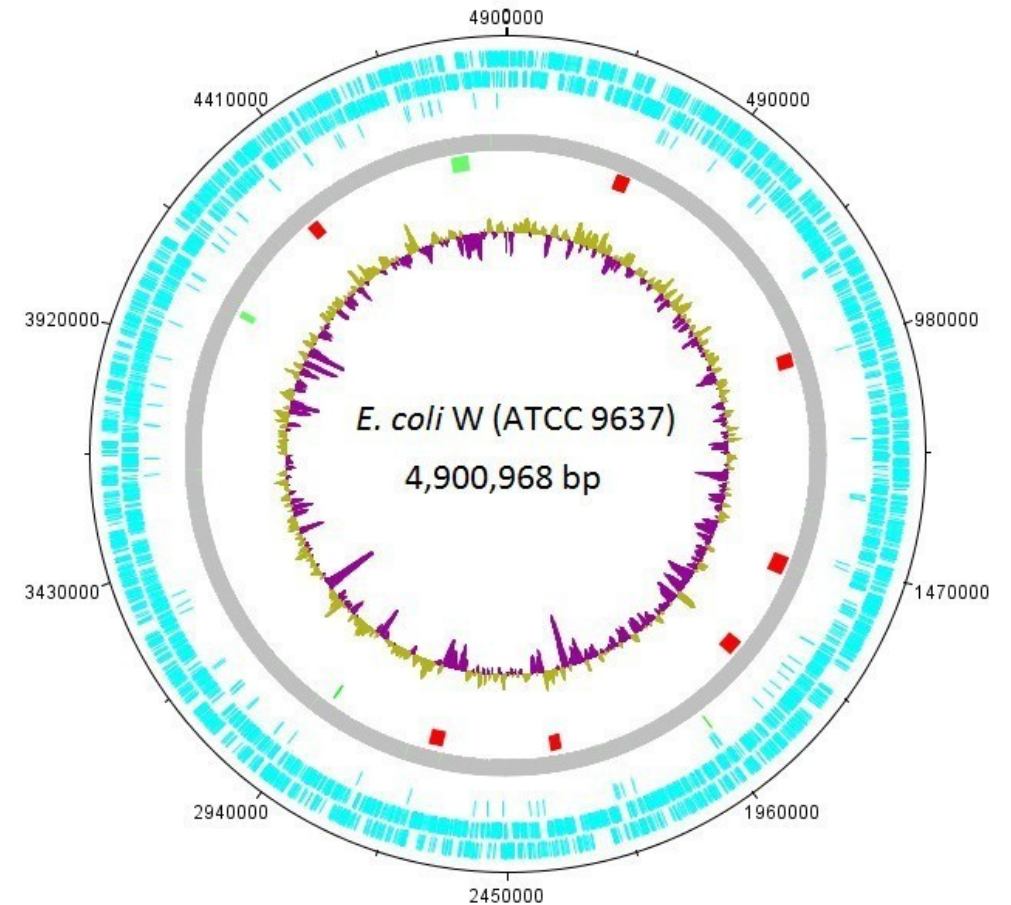


Prokaryotes

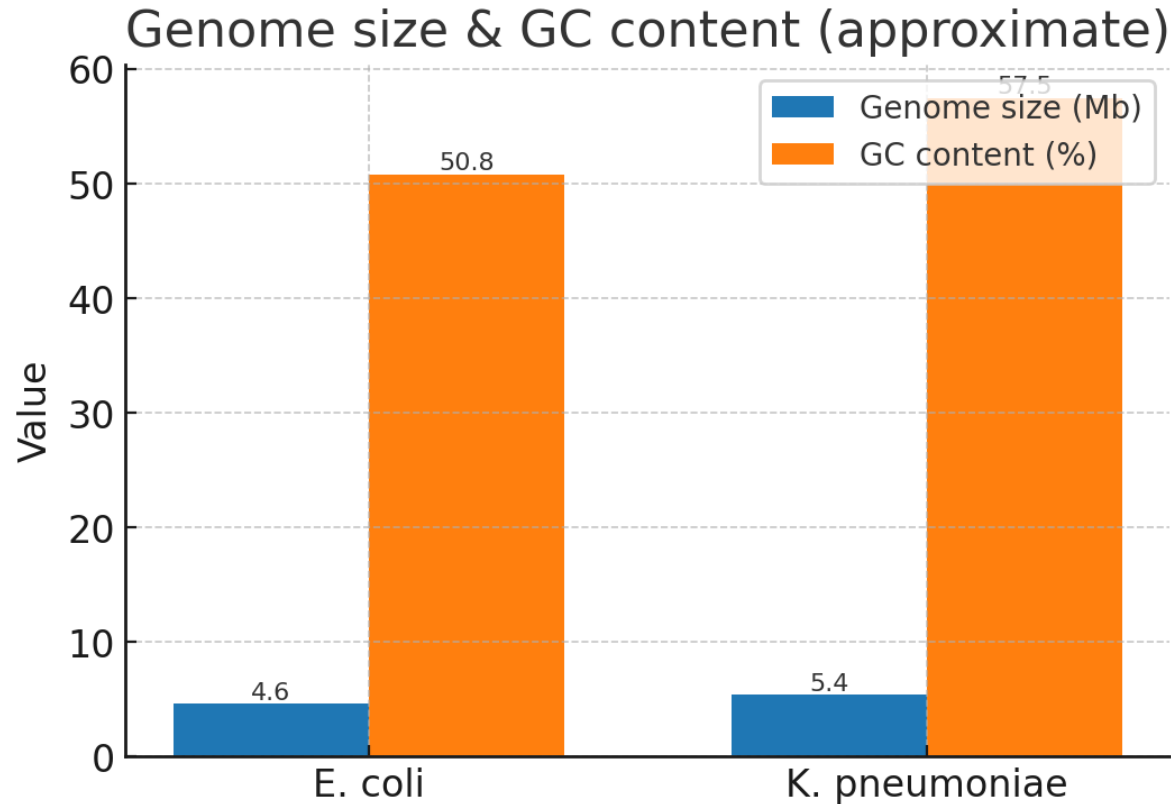
- Prokaryotic genomes generally contain one large circular piece of DNA referred to as a "chromosome" (not a true chromosome in the eukaryotic sense).
- Some bacteria have linear "chromosomes". e.g. spirochete *Borrelia burgdorferi* (the causative agent of Lyme disease)
- Many bacteria have small circular DNA structures called plasmids which can be swapped between neighbors and across bacterial species.

What is a Bacterial Genome?

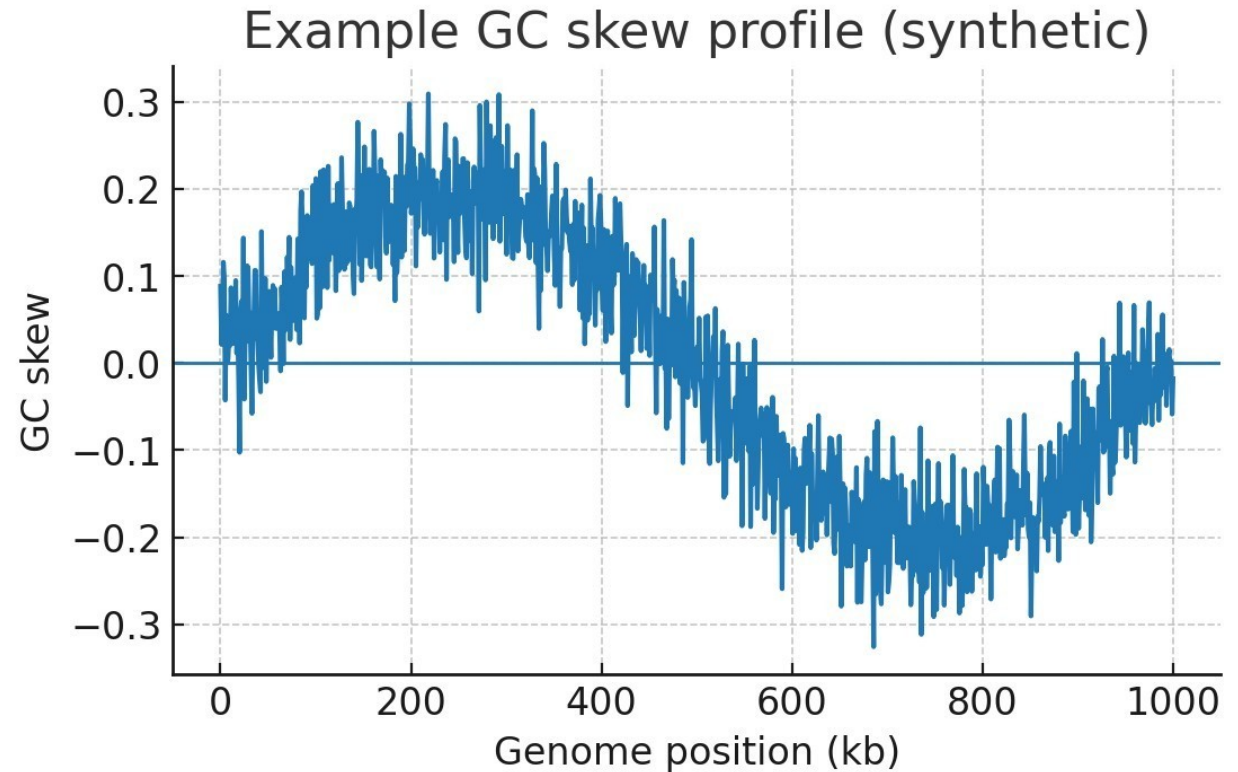
- Circular chromosome in most bacteria (some exceptions linear)
- Size range: typically 0.5 – 10 Mb; *E. coli* ~4.6 Mb, *Klebsiella* ~5.4 Mb
- High gene density: ~85–90% coding sequences
- Accessory elements: plasmids, phages, transposons



Genome Size & GC Content

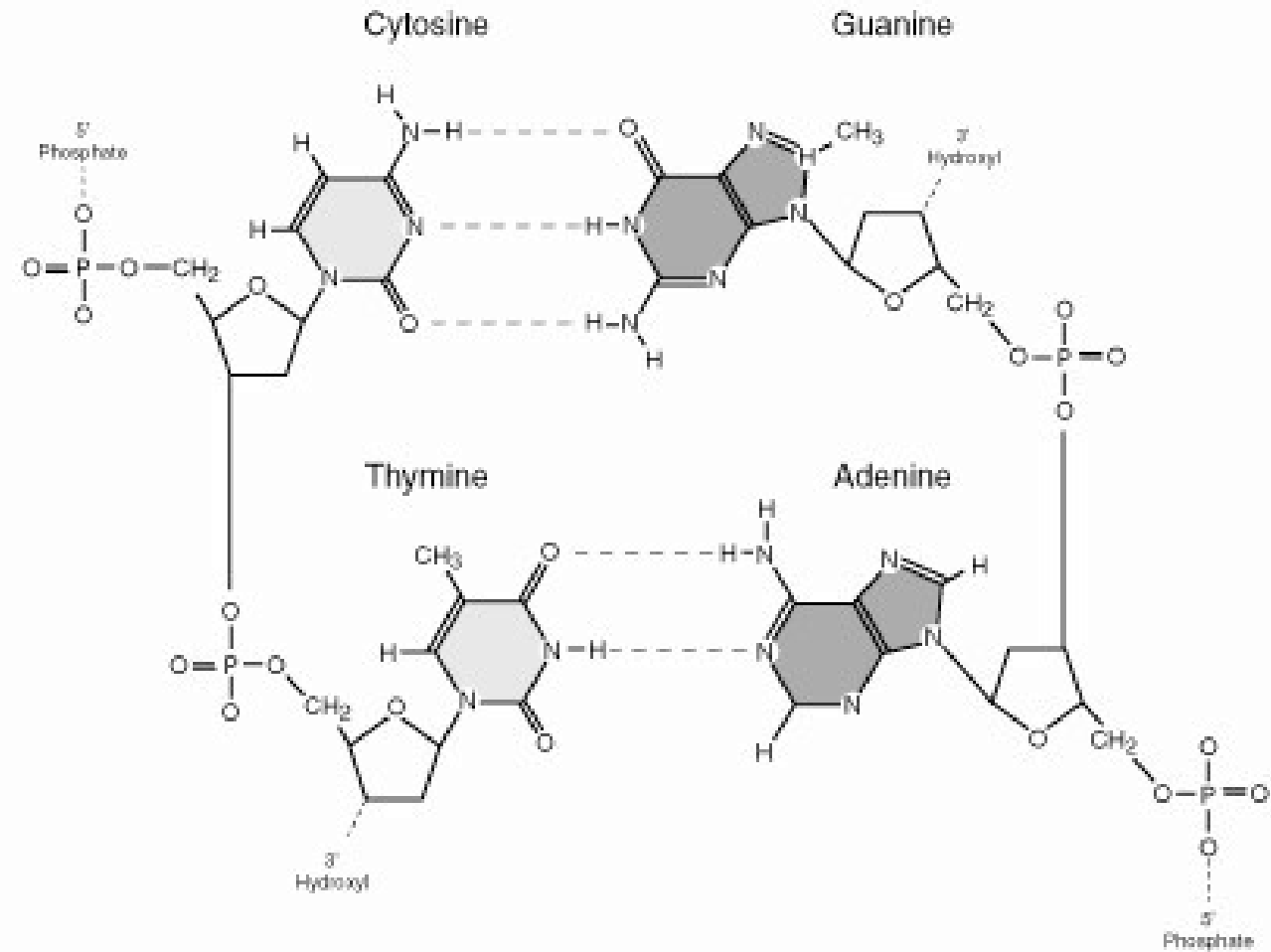
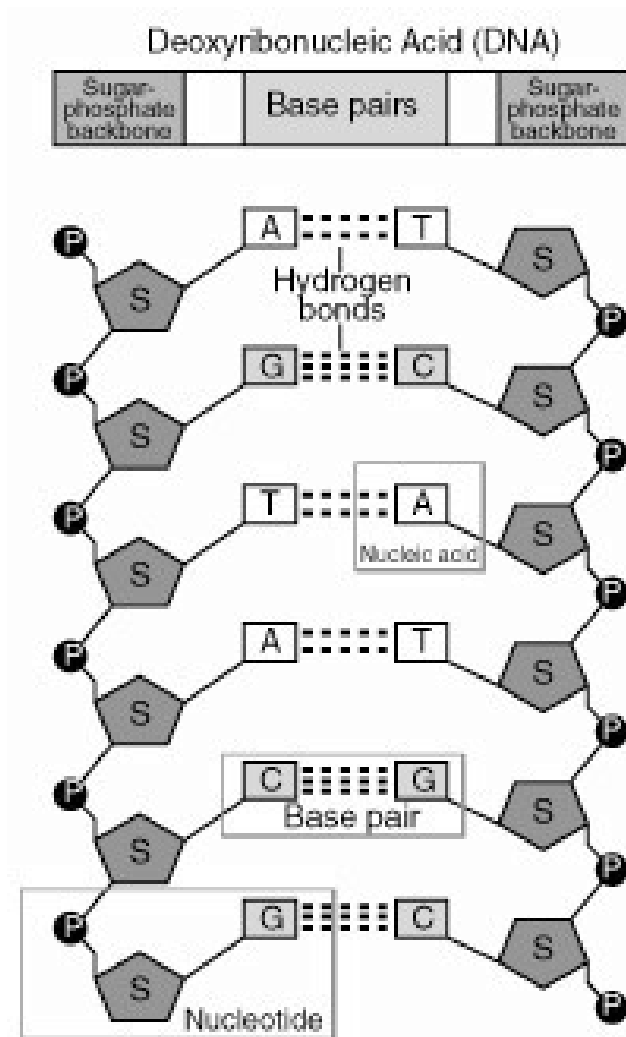


Genome size (Mb) and GC% (approximate)



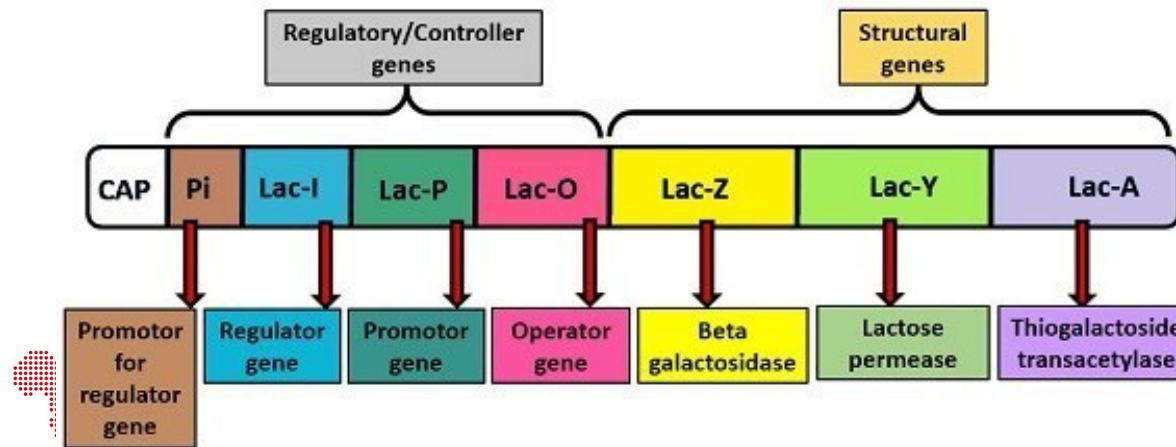
GC skew profile (synthetic example)

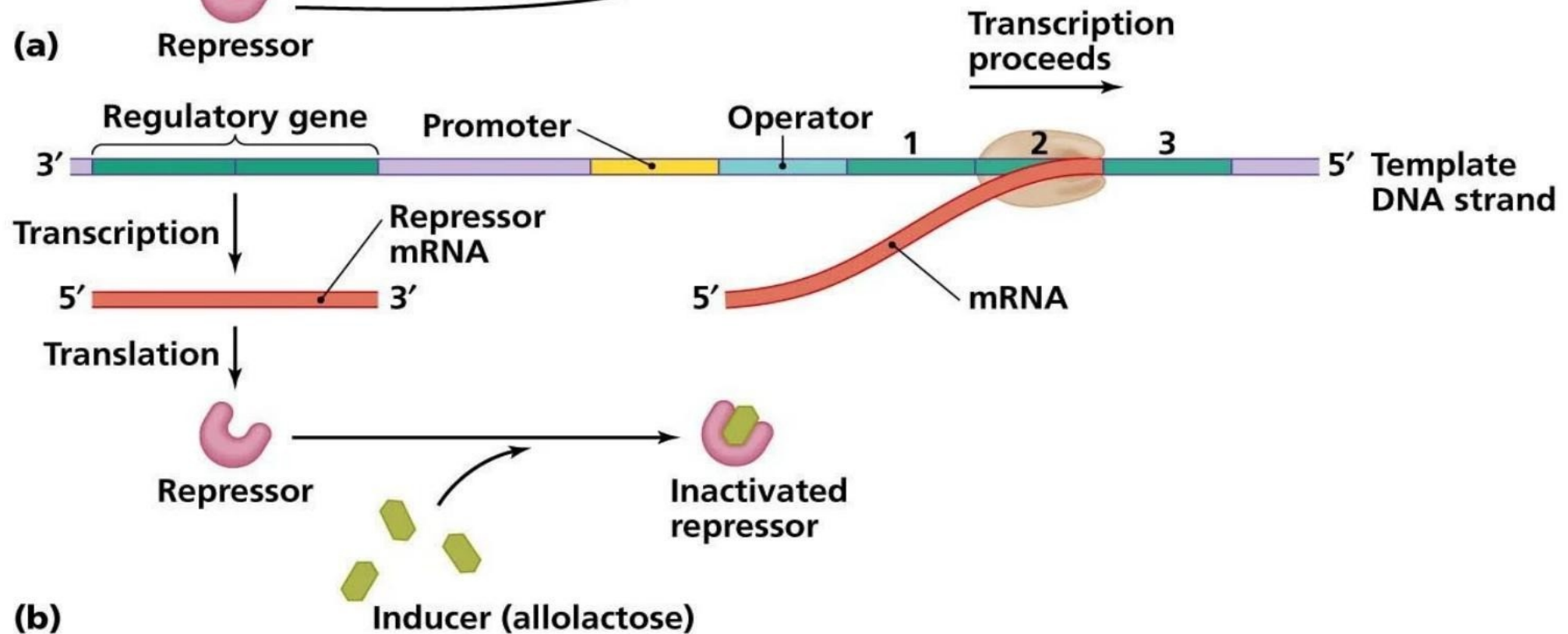
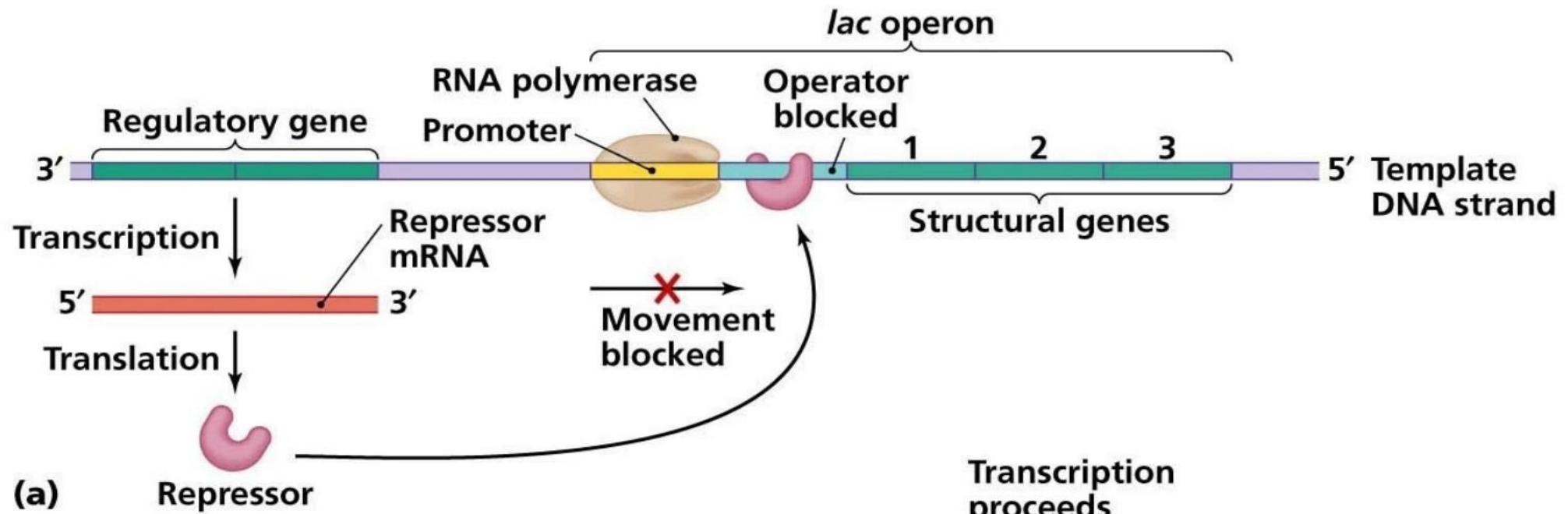
Pairing of bases and nucleotides in DNA



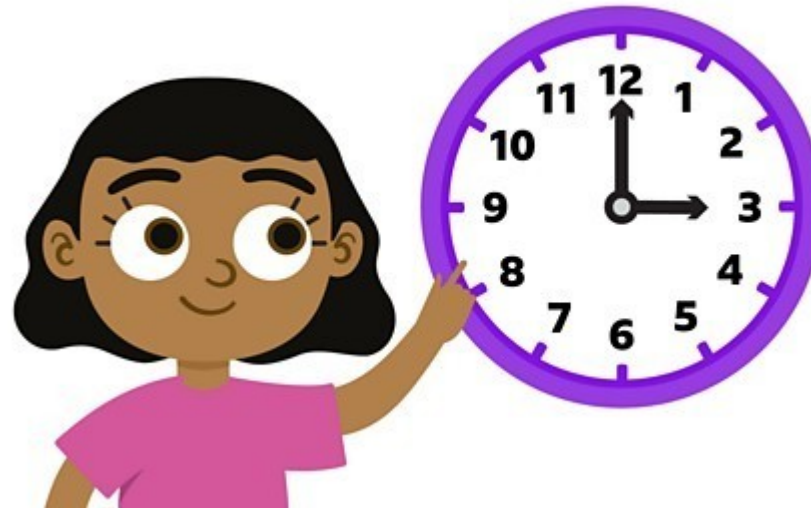
Operon Organization

- An operon is a cluster of genes under the regulation of a single promoter.
- The genes are transcribed together into a single mRNA.
- These genes usually have related functions or are involved in a common metabolic process/pathway.
- A classic example is the **lac operon** in *E. coli*
- The lactose (*lac*) operon includes the genes for β -galactosidase (Z), galactoside permease (Y), and thiogalactoside transacetylase (A).





5 Minutes WhatsApp Break!!!



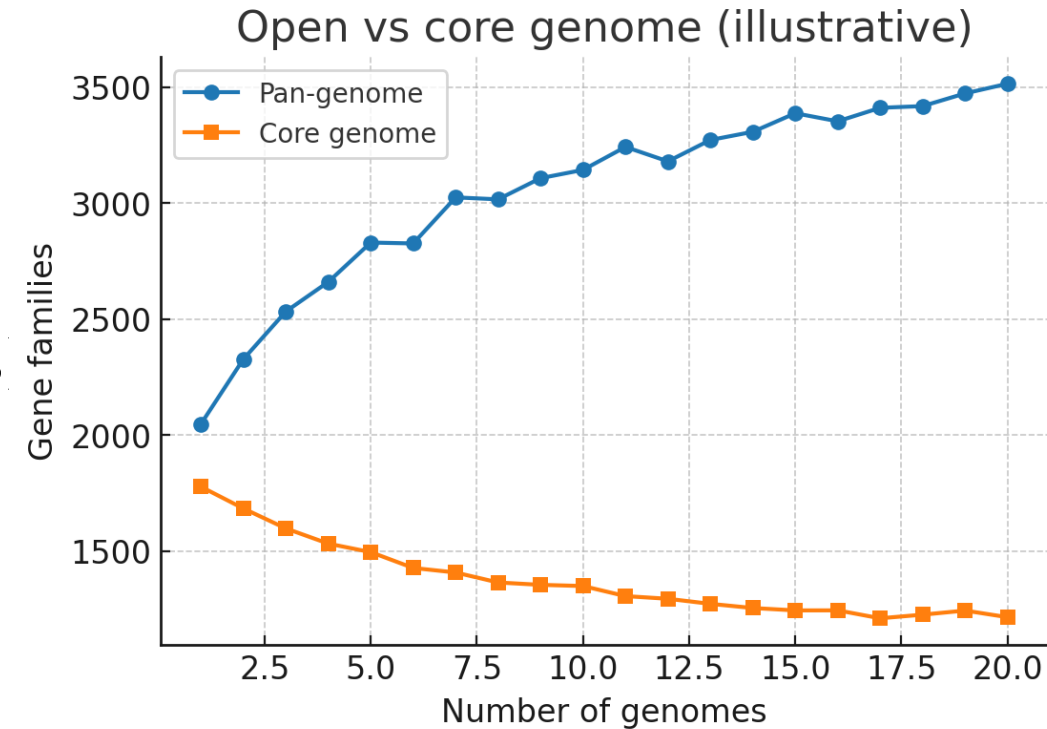
Pan-Genome vs Core Genome

Pan-Genome

- **Definition:** The entire gene repertoire found across all strains of a species.

Components:

- ▶ **Core genome** (shared by all strains)
- ▶ **Accessory genome** (present in some strains; niche adaptation)
- ▶ **Unique genes** (found in only one/few isolates)
- **Dynamic:** Expands as more genomes are sequenced.
- **Example:**
 - AMR genes (e.g., blaCTX-M, blaKPC)
 - Virulence loci (e.g., capsule genes in *Klebsiella*, toxin genes in pathogenic *E. coli*).



Pan-Genome vs Core Genome

Core Genome

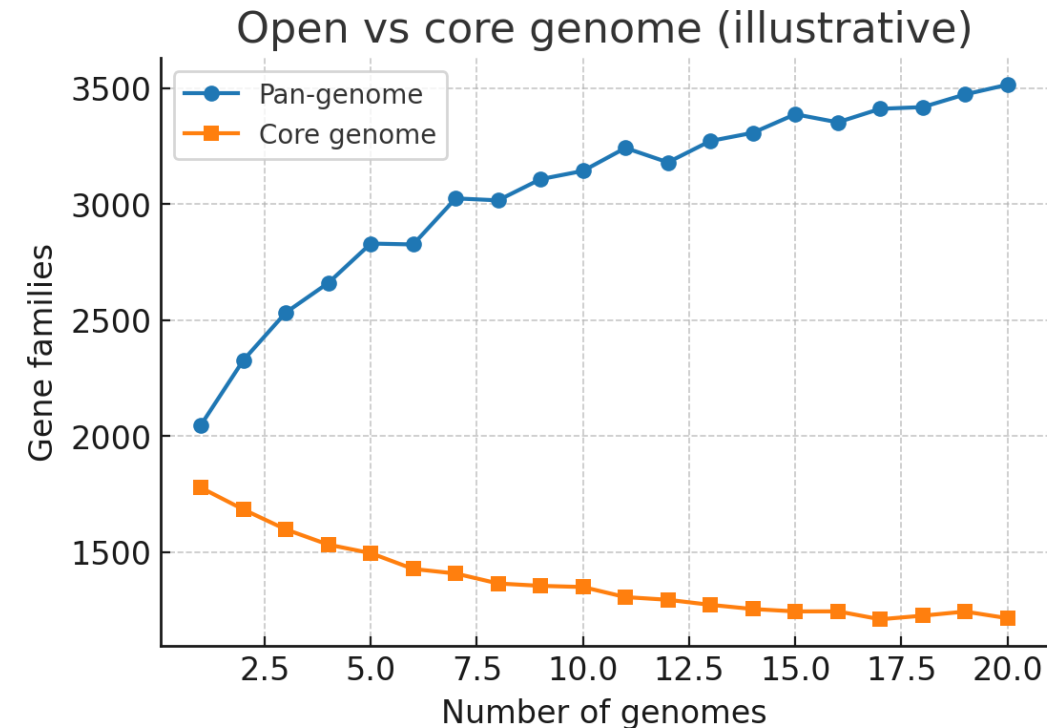
- **Definition:** The set of genes present in all strains of a species.

Represents: Essential functions required

- for survival (e.g., DNA replication, transcription, translation, basic metabolism).
- **Stable:** Highly conserved; changes slowly over evolutionary time.

Example:

- In *E. coli* and *Klebsiella*, the core includes housekeeping genes like *gyrA*, *rpoB*, *recA*.
- Forms the backbone for species identity.

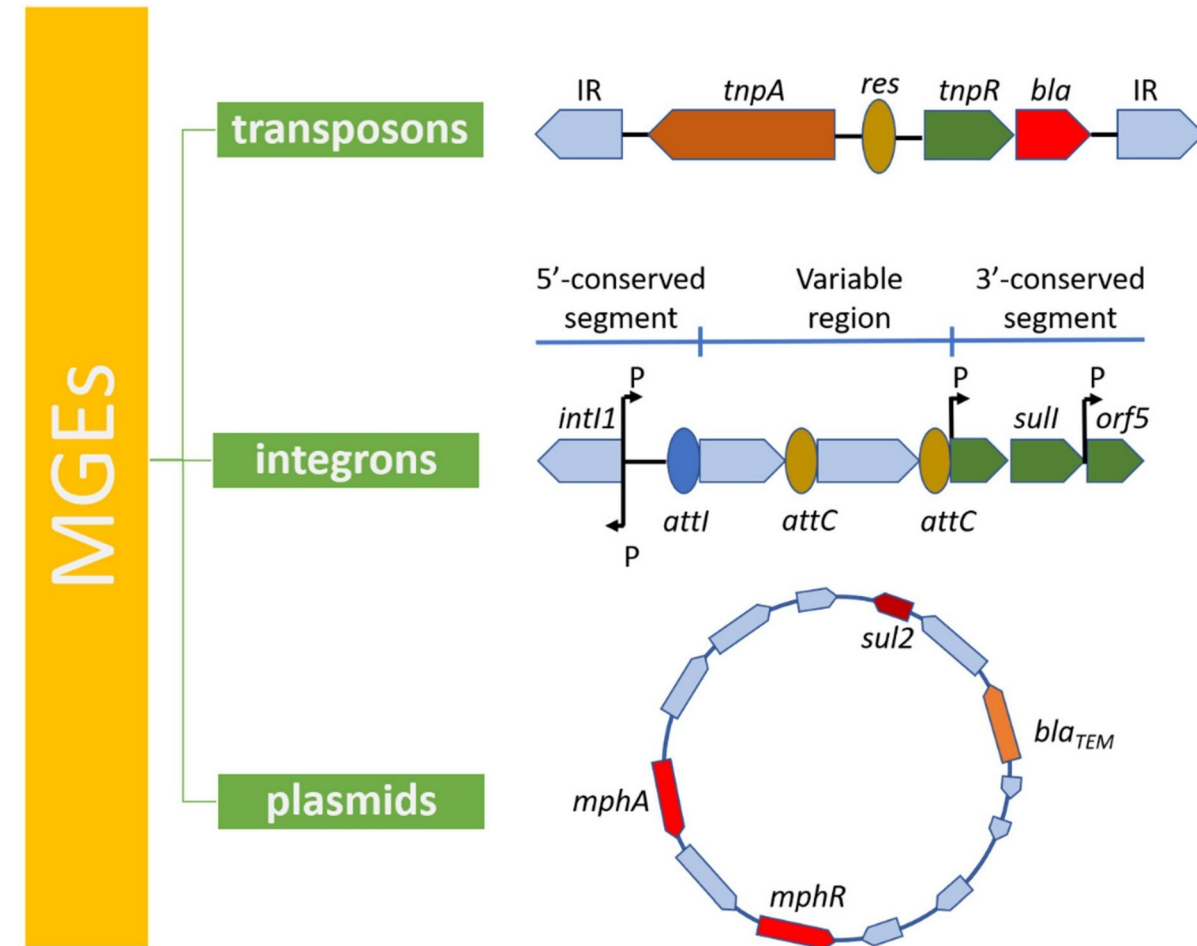


Mobile Genetic Elements

- Segments of DNA that can move within a genome or transfer between different organisms.
- These are sometimes referred to as selfish genetic elements.
- They are a type of genetic material that move around within the genome.

Examples include:

- Plasmids
- Transposons
- Integrations
- Viral agents/ Prophages

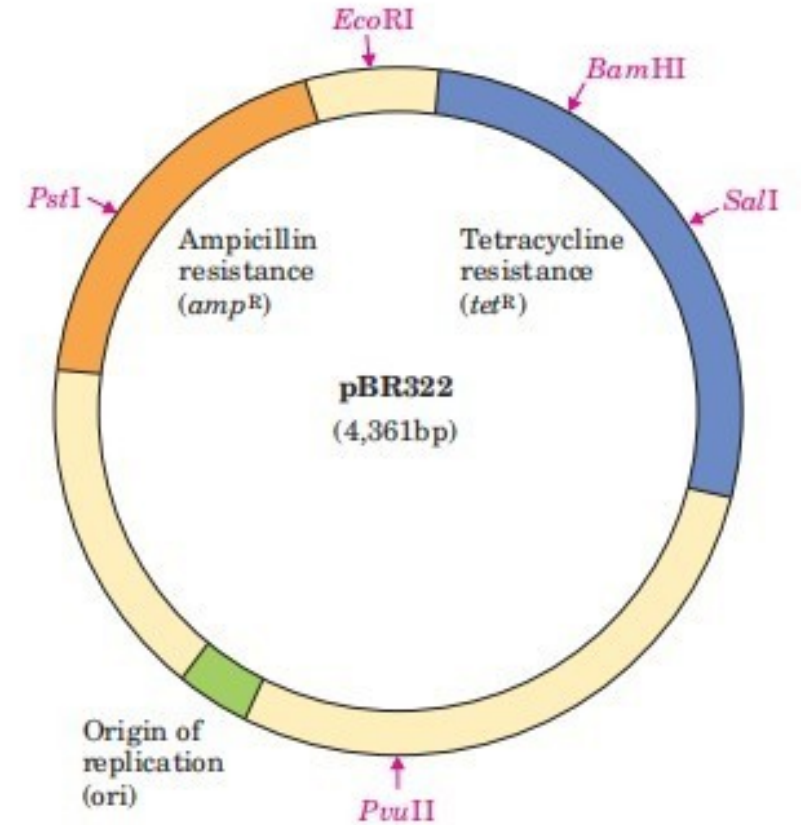


Plasmid

- A **plasmid** is separate from, and can replicate independently of, the chromosomal DNA.
- Plasmid size varies from 1 to over 1,000 (kbp).
- Extra-chromosomal elements inside the bacterial cell.
- Not essential for survival but confer extra advantages to the cell.
- They are most commonly found as circular dsDNA molecules.
- Some cells have none, some have many (20-30), others have multiple copies of the same plasmid.
- They replicate and multiply independently of the chromosome.
- They can be introduced into the cell via transformation.

Plasmids may contain genes that...

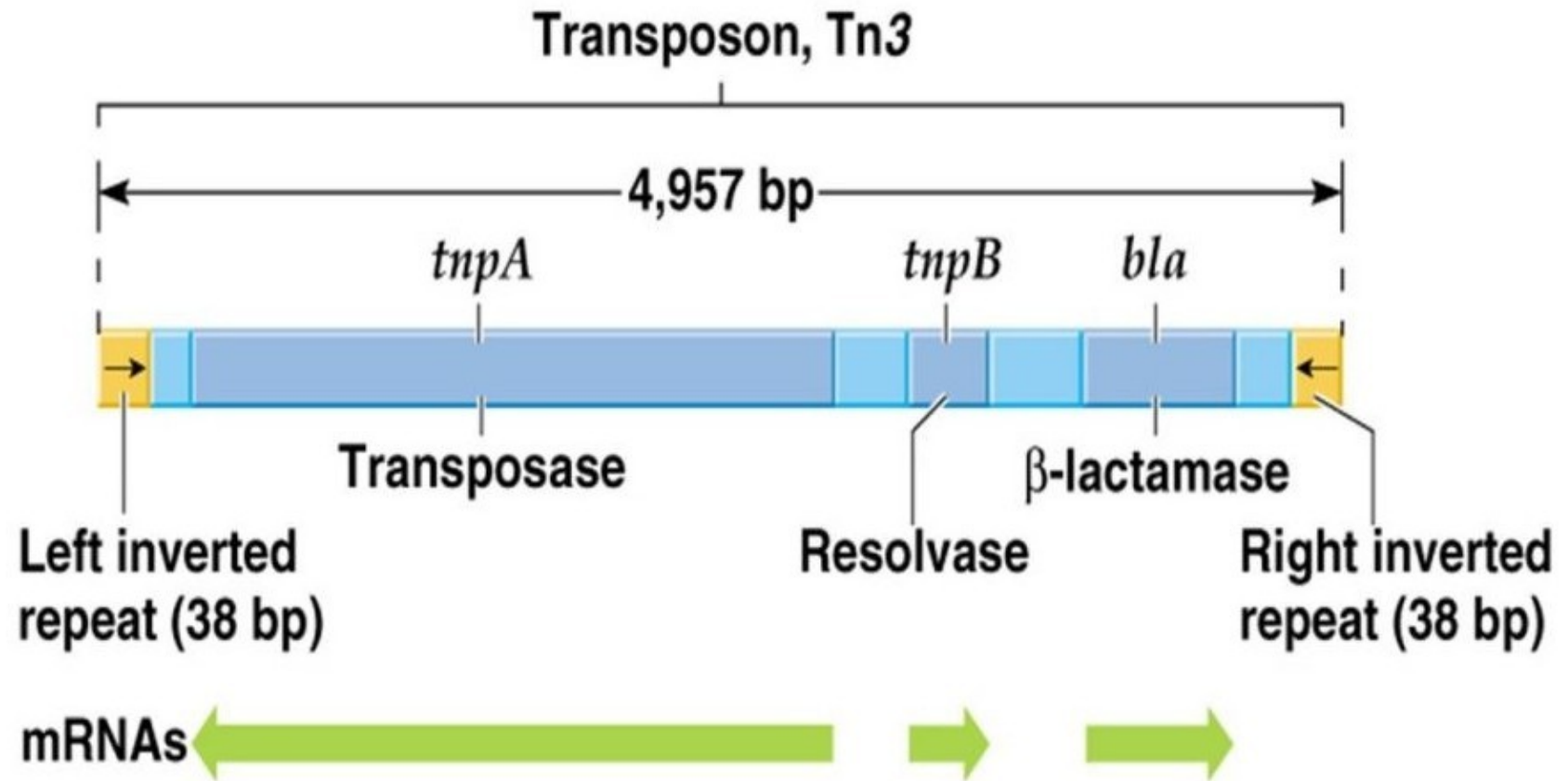
- Code for resistance to antibiotics
- Code for production of toxins
- Code for resistance to heavy metals
- Code for resistance to UV light
- Code for colonisation factors necessary for attachment
- Code for enzymes that allow unique/unusual energy sources



The constructed E. coli plasmid pBR322

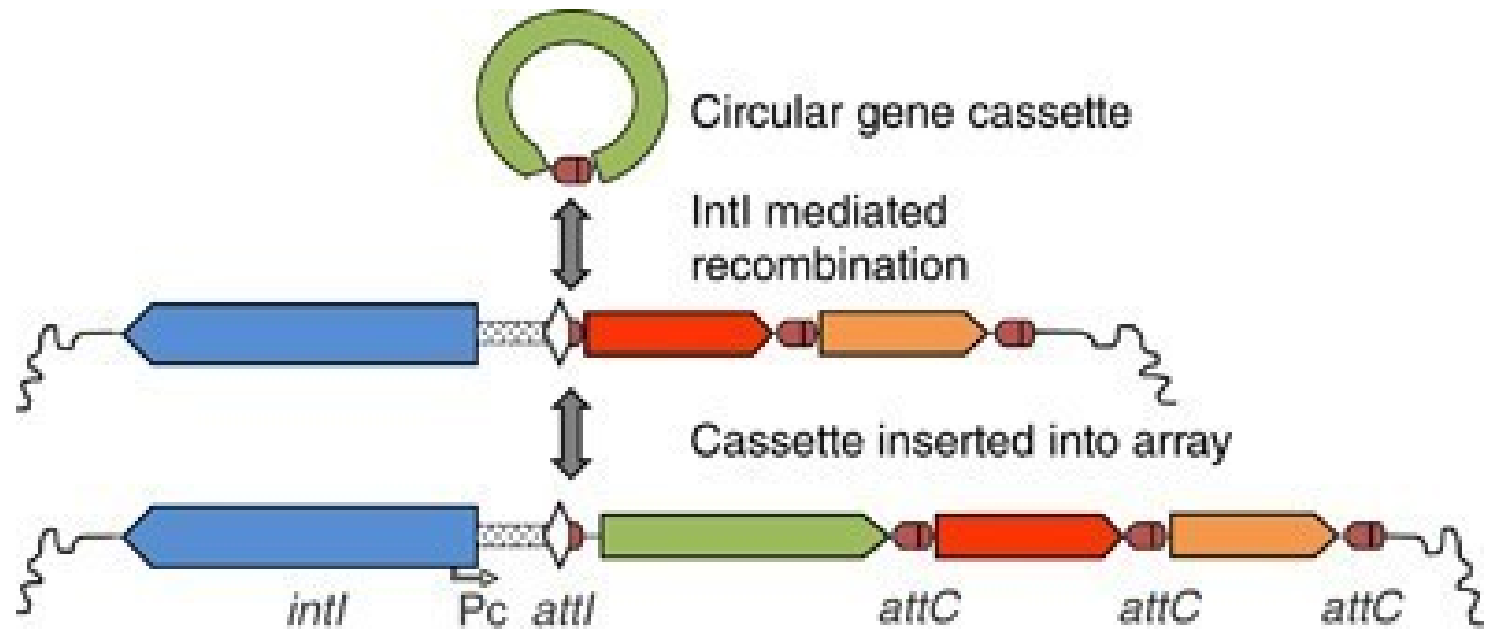
Transposons

- DNA sequences that move from one location in the genome to another by either cutting and pasting themselves or by replicating and inserting copies elsewhere.
- They are also present in eukaryotes (jumping genes).



Integrans

- A gene encoding a site-specific recombinase
- A proximal recombination site where cassettes may be inserted, recognisable by the integrase
- One or more incorporated gene cassettes.
- They capture and express genes (usually AMR genes).
- They may be found as part of other MGEs.



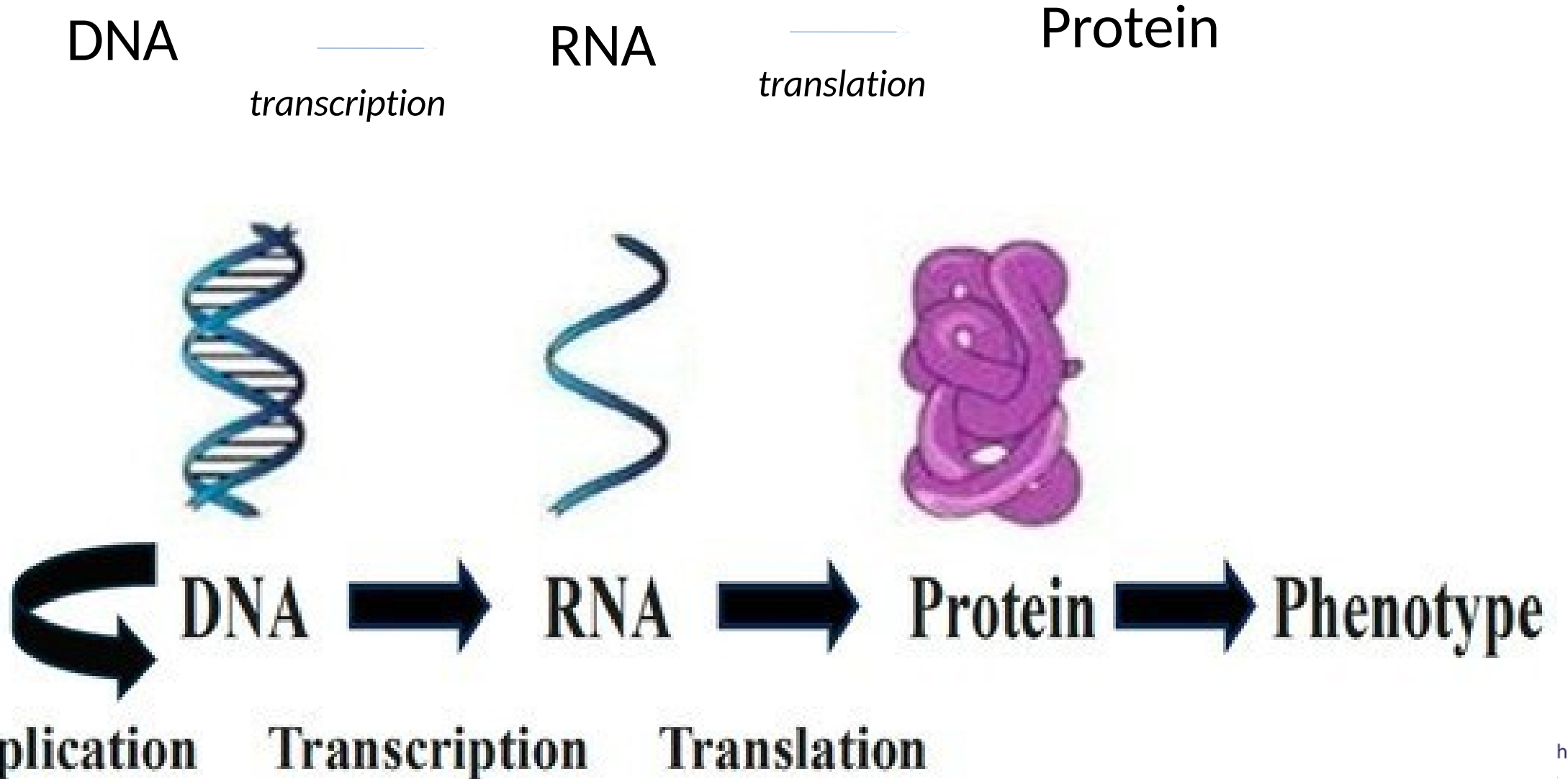
Klebsiella pneumoniae: Genome Features

- Genome: ~5.2–5.8 Mb; GC ~57% (strain-dependent)
- Plasmids common; harbor AMR (e.g., KPC, NDM, OXA-48-like) and virulence factors
- Capsule (K-loci) diversity; hypervirulent lineages (e.g., hvKp) carry siderophores (aerobactin, yersiniabactin)
- High propensity for horizontal gene transfer (HGT); multiple plasmid backbones within hospital clones

Escherichia coli: Genome Features

- Genome: ~4.5–5.5 Mb; GC ~50–51% (strain-dependent)
- Pathotypes (ExPEC, UPEC, EHEC, EPEC, ETEC, EAEC) driven by distinct virulence loci
- Plasmids encode toxins, fimbriae, and AMR determinants; integrons and IS elements are common
- Lab strains (e.g., K-12) vs clinical isolates: notable differences in islands and prophages

Central dogma of molecular biology



One grammar used in genomics: The Genetic Code maps information in DNA (RNA) to protein

		Second base of codon										
		U		C		A		G				
First base of codon	U	UUU	Phenylalanine Phe	UCU	Serine Ser	UAU	Tyrosine Tyr	UGU	Cysteine Cys	Third base of codon	U	
		UUC		UCC		UAC		UGC			C	
		UUA	Leucine Leu	UCA		UAA	STOP	UGA	STOP		A	
		UUG		UCG		UAG		UGG	Tryptophan Trp		G	
	C	CUU	Leucine Leu	CCU	Proline Pro	CAU	Histidine His	CGU	Arginine Arg		U	
		CUC		CCC		CAC		CGC				C
		CUA		CCA		CAA	Glutamine Gin	CGA				A
		CUG		CCG		CAG		CGG				G
	A	AUU	Isoleucine Ile	ACU	Threonine Thr	AAU	Asparagine Asn	AGU	Serine Ser		U	
		AUC		ACC		AAC		AGC			C	
		AUA		ACA		AAA	Lysine Lys	AGA	Arginine Arg		A	
		AUG	Methionine Met	ACG		AAG		AGG			G	
	G	GUU	Valine Val	GCU	Alanine Ala	GAU	Aspartic acid Asp	GGU	Glycine Gly		U	
		GUC		GCC		GAC		GGC				C
		GUA		GCA		GAA	Glutamic acid Glu	GGA				A
		GUG		GCG		GAG		GGG				G
		codons										

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25 words are needed to code for the 20 amino acids and the start and stop sites

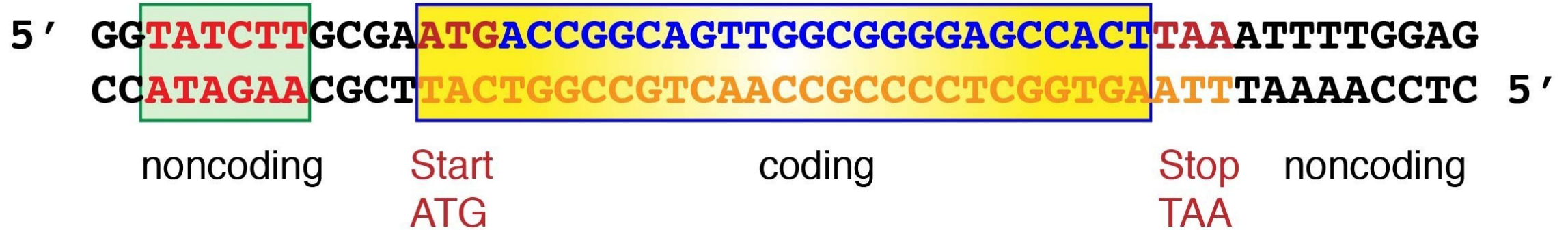
The Triplet Code allows for 64 codons to be coded

=> Degeneracy of the genetic code

Gene structure in bacteria

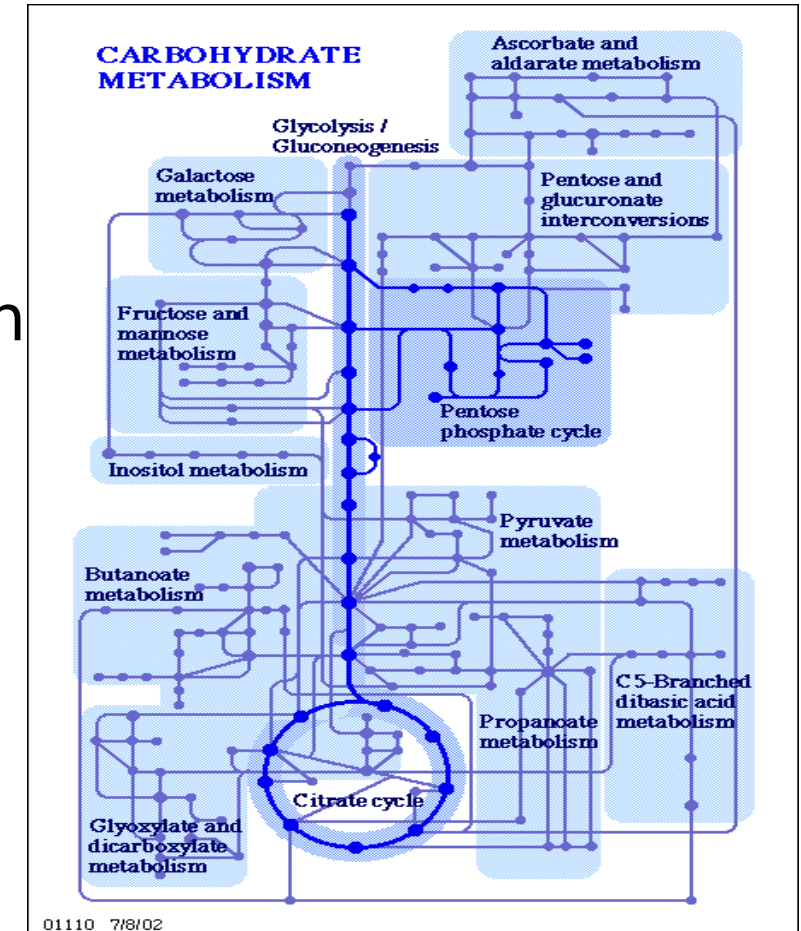
Promoter -10

Open reading frame (ORF)



Predicting functions of candidate protein-coding genes

- Has this sequence been seen before? – Match to sequence database
- “Guilt” by association:
 - Is this sequence similar to a known protein in another species?
 - Is the expression pattern similar to that of known genes? E.g. co-expression with genes ribosomal proteins suggests that the encoded protein could have a ribosomal function
- Deduce physiological function within a context of pathways



KEGG (Ogata et al.1999)

Gene Organization & Regulation

- High gene density; short intergenic regions
- Operons & regulons enable coordinated expression
- Global regulators (e.g., CRP, FNR) integrate environmental cues
- Non-coding RNAs and CRISPR-Cas systems modulate genome defense and regulation

Replication Landmarks & Structure

- OriC: DnaA boxes; bidirectional replication forks
- Ter region & Tus/ter sites in some species
- GC/AT skews, gene orientation bias relative to replication
- Macrodomain organization and nucleoid-associated proteins (NAPs)

Horizontal Gene Transfer (HGT)

- Transformation (rare in Enterobacterales), transduction (phages), conjugation (plasmids)
- Integrative and conjugative elements (ICEs) expand accessory genome
- Genomic islands & pathogenicity islands drive niche adaptation

Clinically Relevant Loci (selected)

- Klebsiella: blaKPC, blaNDM, blaOXA-48-like; aerobactin (iuc), yersiniabactin (ybt), capsule (wzi/wzc typing)
- E. coli: ESBLs (blaCTX-M), carbapenemases (KPC/NDM/OXA-48-like); LEE island (EHEC/EPEC), stx phages
- Efflux pumps, porin changes, and target mutations (gyrA, parC) contribute to resistance

Key Takeaways

- Bacterial genomes are compact yet dynamic: core + accessory components
- *E. coli* and *Klebsiella* share architecture but differ in GC%, virulence, and AMR landscapes
- Operons and regulons underpin efficient regulation
- Plasmids and MGEs drive adaptation, AMR, and virulence central to hospital AMR